

12. Which of the following statements is not true about Matasi soil?

- (a) This soil is usually found at altitudes higher than that of Bhata soil and lower than that of Kanhar soil
- (b) This soil is less fertile than Kanhar soil
- (c) Iron content is high in this soil
- (d) Sand content is high in this soil

Chhattisgarh P.C.S. (Pre) 2021

Ans. (a)

Matasi soil is also called red-yellow soil. This soil is spread in most part of Chhattisgarh. This type of soil about 55 to 60% of the area of Chhattisgarh. It is found at higher altitudes and in places with more drainage. Iron oxide is the most abundant element in this soil. The presence of ferrous oxide gives it a red colour, while ferric oxide gives it a yellow colour. It is a prime soil for paddy cultivation.

v. Acidic and Alkali Soil

*The soil pH is a measure of the acidity or alkalinity in the soil- pH below 7 is **acidic** and above 7 is alkaline. Generally, the pH value of crop soil is 6.0-7.0. *Acidic soil is **sedimentary** in nature. It has mixed laterite, ferrous red and another red soil group. *These soils are formed by the humid climate, acidic material and geomorphology. Due to temperature, humid climate and heavy rain, rapid erosion occurs in soil and increases the rate of leaching of bases. Due to high temperature with heavy rainfall, and leaching, acidic soil is formed.

*Factors causing acidity in the soil include leaching (due to heavy rainfall), acidic substances, acid-forming fertilizers and soluble salts, etc.

Alkali Soil –

*Generally, the alkaline soil is found in India in dry and wetland regions. It is formed due to capillary action of cations such as Mg^{+2} , Na^{+} , Ca^{++} , which are deposited on the surface of the soil. This type of soil is mainly distributed in South Punjab, Southern Haryana and West-Rajasthan. *Gypsum ($CaSO_4$) is used to change the alkalinity of the soil and remove the salinity from it.

1. Following are the statements regarding the Usar soil:

- 1. It is reclaimed by adding lime.
- 2. This soil has a pH of more than seven.
- 3. Paddy crops can be grown in this soil.

Select the correct answer from the codes given below:

Codes :

- (a) 1 and 2 are correct (b) 2 and 3 are correct
- (c) only 3 is correct (d) only 1 is correct

U.P. B.E.O. (Pre) 2019

Ans. (b)

Usar soil has a pH of more than 7. There is a severe deficiency of Bacteria, Nitrogen and Zinc in this soil. Pyrites, Gypsum and green manure are used for its treatment. This type of land is suitable for cultivation of tolerant paddy varieties like Narendra Shankar Usar Paddy-1, Usar Paddy-2, etc. The use of lime is for the treatment of acidic soil instead of the treatment of Usar Soil. So, option (b) is the correct answer.

2. A fertile soil, suitable for growing common crops is likely to have a pH value of :

- (a) Three (b) Four
- (c) Six to seven (d) Nine to ten

I.A.S. (Pre) 1993

Ans. (c)

The soil pH is a measure of the acidity or alkalinity in soils. A pH below 7 is acidic and above 7 is alkaline. Soil pH is considered a master variable in soils as it controls many chemical processes that take place. It specifically affects plant nutrient availability by controlling the chemical forms of the nutrients. Most nutrients that plants need can dissolve easily when the pH of the soil solution ranges from 6.0 to 7.0. Thus, option (c) is correct.

3. Which one of the following can be used to make acidic soil cultivable?

- (a) Lime (b) Gypsum
- (c) Calcium (d) Vegetable compost

M.P.P.C.S. (Pre) 2006

Ans. (a)

Generally, lime is used to make acidic soil cultivable. It is notable that excessive alkaline or acidic soil is not good for soil health and it is not used for cultivation of common crops.

4. The solution to the problem of soil salinity and alkalinity is :

- (a) Dry farming method
- (b) Use of Gypsum in the fields
- (c) Afforestation
- (d) Contour cultivation

R.A.S./R.T.S. (Pre) 1996

Ans. (b)

Gypsum is used to change the alkalinity of the soil and remove the salinity from it. It also requires one foot of water for soil leaching.

5. The long term solution of soil salinity and alkalinity is the use of :

- (a) Rock phosphate (b) Gypsum
(c) Manure (d) Urea

R.A.S./R.T.S. (Pre) 1997

Ans. (b)

See the explanation of the above question.

6. In India maximum alkali area is found in the State of—

- (a) Gujarat (b) Haryana
(c) Punjab (d) Uttar Pradesh

U.P.P.C.S. (Mains) 2007

Ans. (d)

Option (d) was the correct answer during the question period. According to the Barrea Land Atlas of India, 2019, the area (in sq. km) of the states affected by Alkalinity/Salinity in the Year 2015-16 is as follows -

State	Area (Sq. km.)
Gujarat	763.52
Haryana	65.62
Punjab	20.66
Uttar Pradesh	2129.61

But this atlas mentions medium and high Alkalinity/Salinity affected areas.

According to the Desertification and Land Degradation Atlas of India 2021, Gujarat (2598828 ha.), Haryana (33640 ha.) and Uttar Pradesh (282913 ha.) are the states affected by salinity in the year 2018-19. According to this, the state most affected by salinity is Gujarat.

7. In India, the largest area of Saline soil is found in the state of –

- (a) Uttar Pradesh (b) Haryana
(c) Gujarat (d) Maharashtra

U.P.P.C.S. (Mains) 2012

Ans. (c)

See the explanation of the above question.

8. The state with the largest area of salt affected soils is:

- (a) Tamil Nadu
(b) Rajasthan

- (c) Gujarat
(d) Andhra Pradesh

U.P.P.C.S. (Mains), 2017

Ans. (c)

See the explanation of the above question.

9. Salinization occurs when the irrigation water accumulated in the soil evaporates, leaving behind salts and minerals. What are the effects of salinization on the irrigated land?

- (a) It greatly increases the crop production
(b) It makes some soils impermeable
(c) It raises the groundwater level
(d) It fills the air spaces in the soil with water

I.A.S. (Pre) 2011

Ans. (b)

Soil salinization is often a problem in arid regions where ions of soluble salt build up in the soil. In these regions, where irrigation is required for plants to grow, the processes of evaporation and transpiration leave the salt behind in the soil. Over several years, the amount of salt grows to the point that it hinders both seed germination and plant growth, reducing crop yields and eventually rendering soil unusable.

10. Given below are two statements, in which one is labelled as Assertion (A) and the other as Reason (R) :

Assertion (A) : Saline soils in India are not appropriate for cultivation of crops.

Reason (R) : Saline soils have high proportion of salt, largely because of dry climate and poor drainage.

Select the correct answer using the code given below :

Code :

- (a) Both (A) and (R) are true, but (R) is not correct explanation of (A).
(b) (A) is false but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) (A) is true but (R) is false.

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

In India, saline soils are found to have an excess of Sodium and Potassium a high Magnesium ratio and deficiency of lime, hence this soil is not suitable for agriculture. Therefore, statement (A) is true. Saline soil contains an excess amount of salt, primarily due to the dry climate and poor drainage. Therefore, statement (R) is responsible for assertion (A). Hence option (c) is the correct answer.